

Problem Set 1

Applied Stats/Quant Methods 1

Due: October 8, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in **R**, please include the code you used to get your answers. Please also include the **.R** file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 8, 2026. No late assignments will be accepted.

Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

```
1 y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113, 112, 98,  
      80, 97, 95, 111, 114, 89, 95, 126, 98)
```

1. Find a 90% confidence interval for the average student IQ in the school (As a hint, you first need the mean and SD to create a CI):

Answer:

First, find mean and standard deviation of the sample

```

1 mean_y <- mean(y)
2 s_y <- sd(y)

```

After the calculation, we found that sample mean $\bar{x} = 98.44$, and standard deviation $s = 13.09$. Then we can find standard error using R.

```

1 se <- s_y / sqrt(length(y))

```

We get standard error $se = 2.62$, $df = n - 1 = 25 - 1 = 24$. For a 90% confidence interval, $\alpha = 0.10$. Since this is a two-sided confidence interval, $\alpha/2 = 0.05$. With $df = 24$, the t-value is $t = 1.711$.

```

1 t <- 1.711
2 ci_lower <- mean_y - t*se
3 ci_upper <- mean_y + t*se
4 confident_interval <- c(ci_lower, ci_upper)

```

The 90% confidence interval for the average student IQ in the school is (93.96, 102.92).

We can also check the result with R code.

```

1 ci <- t.test(y, conf.level = 0.90)$conf.int

```

2. Next, the school counselor was curious whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country.

Using the same sample, conduct the appropriate hypothesis test with $\alpha = 0.05$.

Answer:

$$H_0 : \mu \leq 100$$

$$H_A : \mu > 100$$

Using formula:

$$t = \frac{\bar{x} - \mu_0}{se}$$

We can find t in R.

```
1 t <- (mean_y - 100) / se
```

$t = -0.596$, meaning the sample mean is 0.596 standard errors below the hypothesized population mean.

Finally, calculate P-value in R.

```
1 # since it is a right tail test we set lower.tail = FALSE
2 p_value <- pt(q=t, df = length(y)-1, lower.tail = FALSE)
```

$P - value = 0.72$

Since $P\text{-value} > \alpha(0.05)$, we fail to reject the null hypothesis. There is insufficient evidence to conclude that the average student IQ in this school is greater than the average IQ score of 100 among all schools in the country.

Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

State	50 states in US
Y	per capita expenditure on shelters/housing assistance in state
X1	per capita personal income in state
X2	Number of residents per 100,000 that are "financially insecure" in state
X3	Number of people per thousand residing in urban areas in state
Region	1=Northeast, 2= North Central, 3= South, 4=West

Explore the `expenditure` data set and import data into R.

```
1 expenditure <- read.table("https://raw.githubusercontent.com/ASDS-TCD/StatsI_2026/main/datasets/expenditure.txt", header=T)
```

- Please plot the relationships among Y , $X1$, $X2$, and $X3$? What are the correlations among them (you just need to describe the graph and the relationships among them)?
- Please plot the relationship between Y and $Region$? On average, which region has the highest per capita expenditure on housing assistance?
- Please plot the relationship between Y and $X1$? Describe this graph and the relationship. Reproduce the above graph including one more variable $Region$ and display different regions with different types of symbols and colors.